

CodeArena Database Schema

Complete JPA Entity Architecture, Table Definitions, and Relationships

1. System & Architecture Overview

The CodeArena platform utilizes a relational database architecture managed via **Spring Data JPA (Hibernate)**. The system comprises **6 primary tables** and **3 enumerated types** designed to handle user identity, programming problems, automated grading test cases, user code submissions, detailed execution results, and real-time leaderboard rankings.

Entity Class	SQL Table Name	Primary Purpose
User	users	Stores user authentication, emails, password hashes, and system roles.
Problem	problems	Stores coding challenge prompts, markdown descriptions, difficulty, and tags.
TestCase	test_cases	Stores evaluation test cases (hidden & public) linked to specific problems.
Submission	submissions	Tracks code executions submitted by users for specific coding challenges.
SubmissionResult	submission_results	Stores granular test-case level outcomes, stdout, and execution times.
LeaderboardEntry	leaderboard_entries	Maintains aggregated user rankings, problems solved, and accuracy scores.

2. Detailed Table Specifications

Table: users (User Entity)

Core user accounts table storing authentication credentials, unique handles, and system access levels.

Column Name	Data Type	SQL Constraints & Notes
id	BIGINT (PK)	AUTO_INCREMENT / IDENTITY. Primary key.
username	VARCHAR(100)	NOT NULL, UNIQUE. User login handle.
email	VARCHAR(150)	NOT NULL, UNIQUE. User email address for notifications & login.
password_hash	VARCHAR(255)	NOT NULL. BCrypt encoded secret password.
role	VARCHAR(50)	NOT NULL. Enum (UserRole): 'USER' or 'ADMIN'. Defaults to 'USER' via @PrePersist.
created_at	TIMESTAMP	NOT NULL. Account registration timestamp. Defaults to now().

Table: problems (Problem Entity)

Stores competitive coding challenge definitions, full descriptions in Markdown format, and difficulty rankings.

Column Name	Data Type	SQL Constraints & Notes
id	BIGINT (PK)	AUTO_INCREMENT / IDENTITY. Primary key.
title	VARCHAR(255)	NOT NULL. Challenge title (e.g., 'Two Sum').
description_md	TEXT	NOT NULL. Complete problem description, input/output formats in Markdown.
difficulty	VARCHAR(20)	NOT NULL. Enum (DifficultyLevel): 'EASY', 'MEDIUM', or 'HARD'.
tags	VARCHAR(500)	Nullable. Comma-separated or JSON string of topics (e.g., 'Arrays, Hash Table').
created_at	TIMESTAMP	NOT NULL. Problem creation timestamp.

Table: test_cases (TestCase Entity)

Defines standard inputs and expected outputs for evaluating user submissions against specific problems.

Column Name	Data Type	SQL Constraints & Notes
id	BIGINT (PK)	AUTO_INCREMENT / IDENTITY. Primary key.
problem_id	BIGINT (FK)	NOT NULL. Foreign key pointing to problems(id). Many-to-One relationship.
input	TEXT	NOT NULL. Raw stdin input passed to the execution container.
expected_output	TEXT	NOT NULL. Exact expected stdout string required for passing.
is_hidden	BOOLEAN	NOT NULL. DEFAULT false. If true, test case values are hidden from users.

Table: submissions (Submission Entity)

Records all code submissions attempted by users, tracking the source code, target language, and overall verdict.

Column Name	Data Type	SQL Constraints & Notes
id	BIGINT (PK)	AUTO_INCREMENT / IDENTITY. Primary key.
problem_id	BIGINT (FK)	NOT NULL. Foreign key pointing to problems(id). Many-to-One.
user_id	BIGINT (FK)	NOT NULL. Foreign key pointing to users(id). Many-to-One.
language	VARCHAR(50)	NOT NULL. Programming language (e.g., 'PYTHON', 'JAVA', 'CPP').
code	TEXT	NOT NULL. Full source code submitted by the user.
status	VARCHAR(50)	NOT NULL. Enum (SubmissionStatus). Defaults to 'PENDING' via @PrePersist.
submitted_at	TIMESTAMP	NOT NULL. Submission timestamp. Defaults to now().

Table: submission_results (SubmissionResult Entity)

Detailed breakdown of how a submission performed on each individual test case, including runtime performance.

Column Name	Data Type	SQL Constraints & Notes
id	BIGINT (PK)	AUTO_INCREMENT / IDENTITY. Primary key.
submission_id	BIGINT (FK)	NOT NULL. Foreign key pointing to submissions(id). Many-to-One.
test_case_id	BIGINT (FK)	NOT NULL. Foreign key pointing to test_cases(id). Many-to-One.
input_data	TEXT	Nullable. Snapshot of the test case input.
expected_output	TEXT	Nullable. Snapshot of the target output.
actual_output	TEXT	Nullable. Actual stdout produced during execution.
status	VARCHAR(50)	NOT NULL. Verdict for this specific test case.
passed	BOOLEAN	NOT NULL. DEFAULT false. True if actual_output matches expected_output.
error_message	TEXT	Nullable. Stderr / compilation / runtime exception trace.
exec_time_ms	BIGINT	Nullable. Execution duration in milliseconds.
executed_at	TIMESTAMP	NOT NULL. Timestamp of evaluation completion.

Table: `leaderboard_entries` (LeaderboardEntry Entity)

Aggregated performance metrics and overall leaderboard ranking per user.

Column Name	Data Type	SQL Constraints & Notes
<code>id</code>	BIGINT (PK)	AUTO_INCREMENT / IDENTITY. Primary key.
<code>user_id</code>	BIGINT (FK)	NOT NULL, UNIQUE. Foreign key pointing to <code>users(id)</code> . One-to-One relationship.
<code>problems_solved</code>	INTEGER	NOT NULL. Total count of distinct problems successfully solved. DEFAULT 0.
<code>accuracy</code>	DOUBLE	Nullable. Overall submission pass rate (percentage). DEFAULT 0.0.
<code>rank</code>	INTEGER	Nullable. Calculated numeric position on the global leaderboard.

3. Enumerated Types (Enums)

The application utilizes strongly-typed Java Enums persisted as VARCHAR strings within the database using `@Enumerated(EnumType.STRING)`.

Enum Name	Persisted Values	Usage Context
UserRole	USER ADMIN	Determines authorization privileges across the platform and admin panel access.
DifficultyLevel	EASY MEDIUM HARD	Categorizes coding challenges by algorithmic complexity.
SubmissionStatus	PENDING RUNNING ACCEPTED WRONG_ANSWER TIME_LIMIT_EXCEEDED MEMORY_LIMIT_EXCEEDED RUNTIME_ERROR COMPILATION_ERROR	Tracks real-time evaluation states and final verdicts for submissions.

4. Entity Relationships & Foreign Keys

Below is the complete relational mapping maintained between the tables:

Source Entity (Table)	Relationship	Target Entity (Table)	Foreign Key Specification
Problem (problems)	1 : N (One-to-Many)	TestCase (test_cases)	test_cases.problem_id → problems.id
Problem (problems)	1 : N (One-to-Many)	Submission (submissions)	submissions.problem_id → problems.id
User (users)	1 : N (One-to-Many)	Submission (submissions)	submissions.user_id → users.id
Submission (submissions)	1 : N (One-to-Many)	SubmissionResult (submission_results)	submission_results.submission_id → submissions.id <i>CascadeType.ALL, FetchType.EAGER</i>
TestCase (test_cases)	1 : N (One-to-Many)	SubmissionResult (submission_results)	submission_results.test_case_id → test_cases.id
User (users)	1 : 1 (One-to-One)	LeaderboardEntry (leaderboard_entries)	leaderboard_entries.user_id → users.id (UNIQUE)